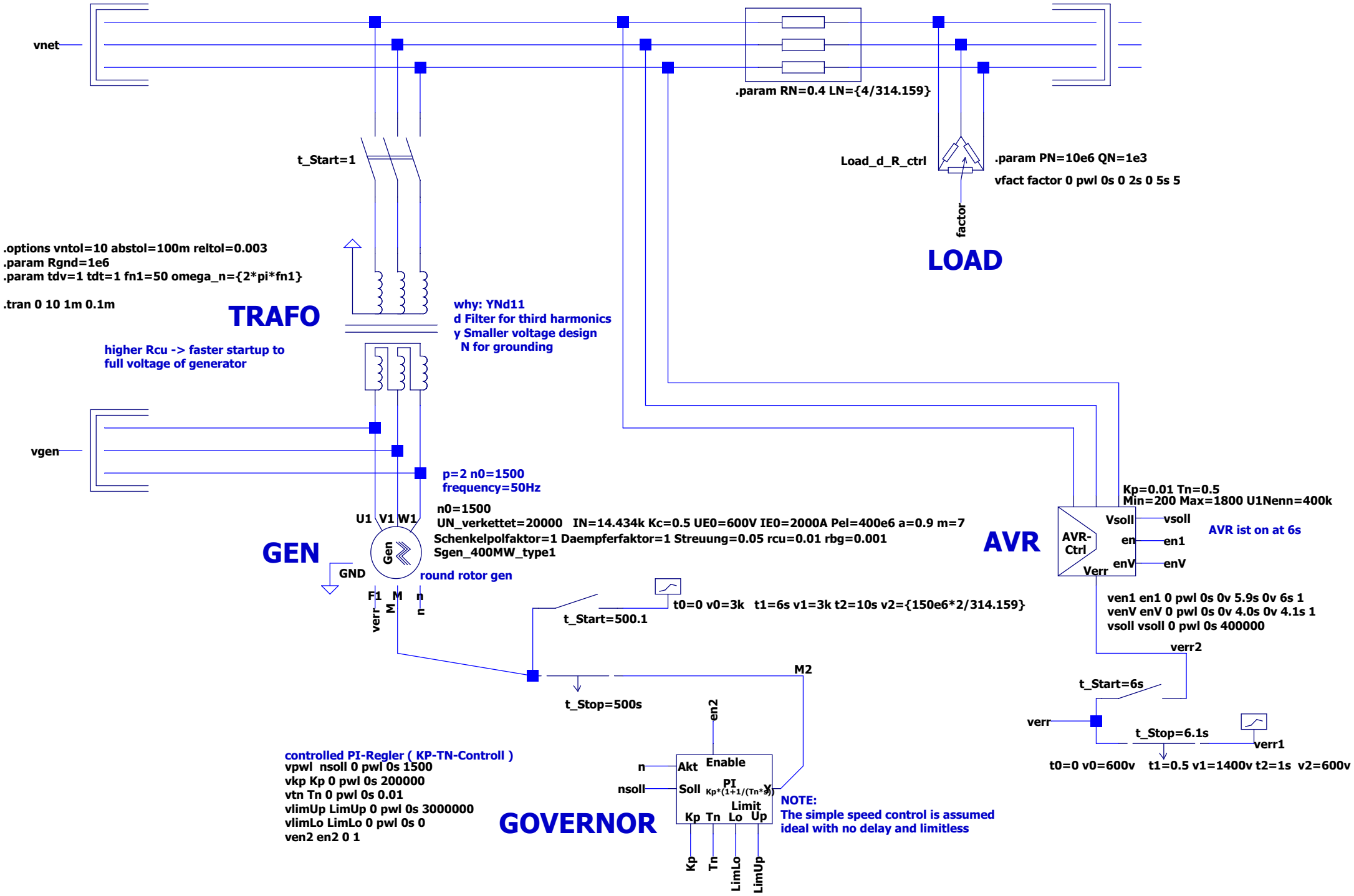
vnet
.param RN=0.4 LN={4/314.159}
t_Start=1
Load_d_R_ctrl
.param PN=10e6 QN=1e3
vfact factor 0 pwl 0s 0 2s 0 5s 5
factor
LOAD
.options vntol=10 abstol=100m reltol=0.003
.param Rgnd=1e6
.param tdv=1 tdt=1 fn1=50 omega_n={2*pi*fn1}
.tran 0 10 1m 0.1m
TRAFO
why: YNd11
d Filter for third harmonics
y Smaller voltage design
N for grounding
higher Rcu -> faster startup to
full voltage of generator
vgen
p=2 n0=1500
frequency=50Hz
n0=1500
UN_verkettet=20000 IN=14.434k Kc=0.5 UE0=600V IE0=2000A Pel=400e6 a=0.9 m=7
Schenkelpolfaktor=1 Daempferfaktor=1 Streuung=0.05 rcu=0.01 rbg=0.001
Sgen_400MW_type1
U1 V1 W1
GEN
Gen
GND
round rotor gen
F1 M n
verr M n
Kp=0.01 Tn=0.5
Min=200 Max=1800 U1Nenn=400k
AVR
AVR-Ctrl
Vsoll
en
Verr
vsoll
en1
enV
AVR ist on at 6s
t0=0 v0=3k t1=6s v1=3k t2=10s v2={150e6*2/314.159}
t_Start=500.1
ven1 en1 0 pwl 0s 0v 5.9s 0v 6s 1
venV enV 0 pwl 0s 0v 4.0s 0v 4.1s 1
vsoll vsoll 0 pwl 0s 400000
verr2
t_Start=6s
verr
t_Stop=6.1s
verr1
t0=0 v0=600v t1=0.5 v1=1400v t2=1s v2=600v
M2
t_Stop=500s
en2
controlled PI-Regler (KP-TN-Controll)
vpwl nsoll 0 pwl 0s 1500
vkp Kp 0 pwl 0s 200000
vtn Tn 0 pwl 0s 0.01
vlimUp LimUp 0 pwl 0s 3000000
vlimLo LimLo 0 pwl 0s 0
ven2 en2 0 1
GOVERNOR
n
nsoll
Akt
Soll
Enable
PI
Kp*(1+1/(Tn*s))
Y
Limit
Kp Tn Lo Up
Kp
Tn
LimLo
LimUp
NOTE:
The simple speed control is assumed
ideal with no delay and limitless